

SQL queries [Basic]

Step 1: Database & Table Basics

```
1  -- Show existing databases
2  SHOW DATABASES;
3
4  -- Create a new database
5  CREATE DATABASE schooldb;
6
7  -- Use the database
8  USE schooldb;
9
10 -- Create a student table
11 CREATE TABLE student (
12     id INT AUTO_INCREMENT PRIMARY KEY,
13     name VARCHAR(50),
14     email VARCHAR(100),
15     age INT
16 );
17
```

Step 2: Insert Data

```
1  -- Insert a single record
2  INSERT INTO student (name, email, age)
3  VALUES ('Alice', 'alice@example.com', 22);
4
5  -- Insert multiple records
6  INSERT INTO student (name, email, age)
7  VALUES
8  ('Bob', 'bob@example.com', 20),
9  ('Carol', 'carol@example.com', 23),
10 ('David', 'david@example.com', 21);
11
```

Step 3: Select / Retrieve Data

```
1  -- Select all columns
2  SELECT * FROM student;
3
4  -- Select specific columns
5  SELECT name, age FROM student;
6
7  -- Use WHERE clause
8  SELECT * FROM student WHERE age > 21;
9
10 -- Use ORDER BY
11 SELECT * FROM student ORDER BY age ASC;
12
13 -- Use LIMIT
14 SELECT * FROM student LIMIT 2;
15
16 -- Use DISTINCT
17 SELECT DISTINCT age FROM student;
```

Step 4: Update & Delete

```

1  -- Update a record
2  UPDATE student SET age = 24 WHERE name = 'Alice';
3
4  -- Delete a record
5  DELETE FROM student WHERE name = 'Bob';
6

```

Step 5: Filtering with WHERE

```

1  -- Use comparison operators
2  SELECT * FROM student WHERE age <= 22;
3
4  -- Use BETWEEN
5  SELECT * FROM student WHERE age BETWEEN 20 AND 22;
6
7  -- Use LIKE for pattern matching
8  SELECT * FROM student WHERE name LIKE 'C%'; -- names starting with 'C'
9
10 -- Use IN
11 SELECT * FROM student WHERE age IN (21, 23);
12

```

Step 6: Aggregate Functions

```

1  -- Get total number of students
2  SELECT COUNT(*) FROM student;
3
4  -- Get average age
5  SELECT AVG(age) FROM student;
6
7  -- Get oldest student
8  SELECT MAX(age) FROM student;
9
10 -- Get youngest student
11 SELECT MIN(age) FROM student;
12
13 -- Sum (not always meaningful for age, but for demo)
14 SELECT SUM(age) FROM student;
15

```

Step 7: GROUP BY and HAVING

```

1  -- Count how many students of each age
2  SELECT age, COUNT(*) FROM student GROUP BY age;
3
4  -- Use HAVING (filter after GROUP BY)
5  SELECT age, COUNT(*) FROM student GROUP BY age HAVING COUNT(*) > 1;

```

Step 8: Table Alterations

```
1 -- Add a new column
2 ALTER TABLE student ADD gender VARCHAR(10);
3
4 -- Rename a column (MySQL 8+)
5 ALTER TABLE student RENAME COLUMN gender TO sex;
6
7 -- Drop a column
8 ALTER TABLE student DROP COLUMN sex;
9
```

Step 9: Dropping Tables and Database

```
1 -- Drop a table
2 DROP TABLE enrollment;
3
4 -- Drop a database
5 DROP DATABASE schooldb;
6
```